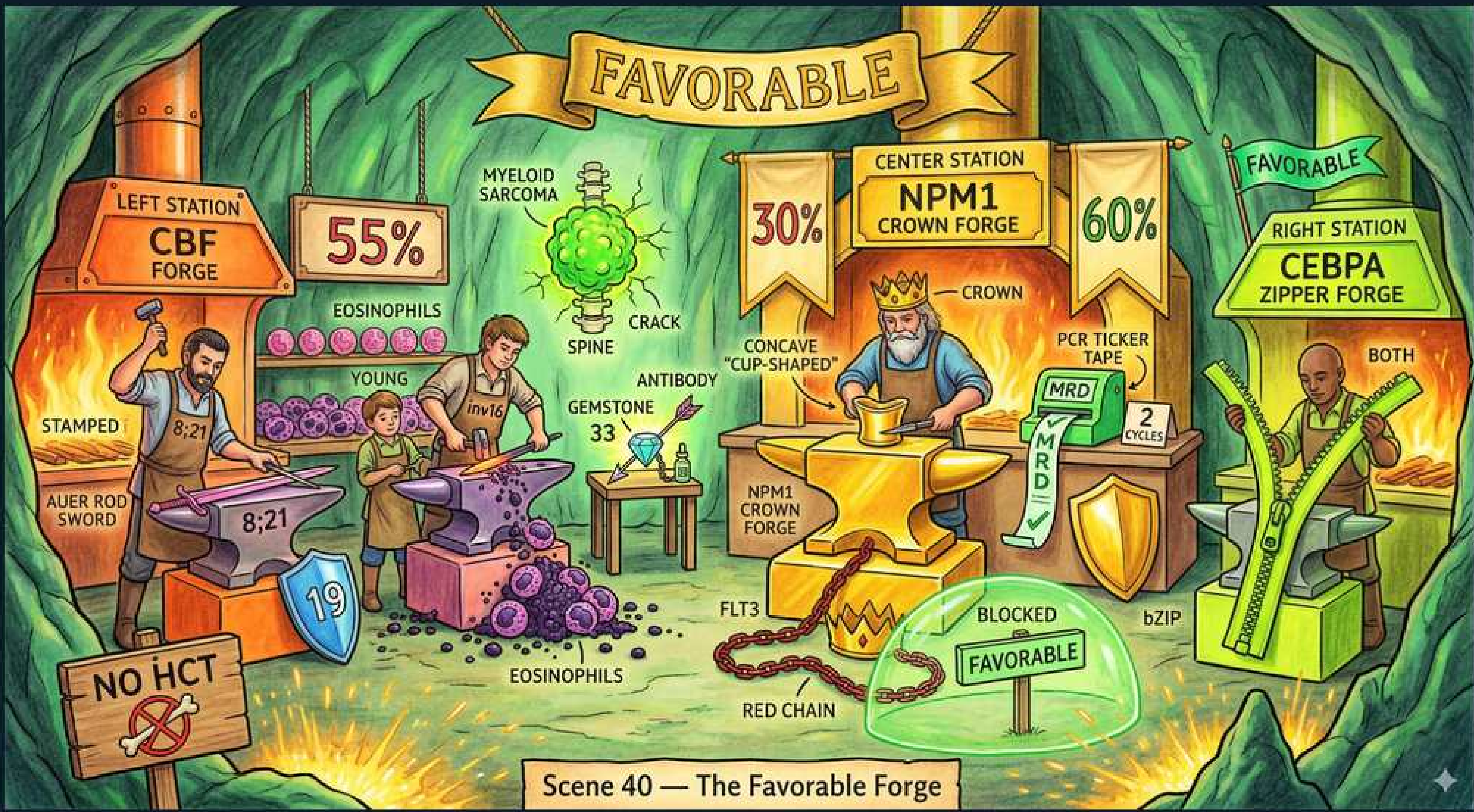
AML — Favorable Genetics (Classification)



1. FAVORABLE banner

WHAT YOU SEE

A large golden FAVORABLE banner draped over the whole forge — the bright banner marks the best-survival ELN group cured by chemo without CR1 transplant.

WHAT IT ENCODES

Per ELN 2022, favorable-risk AML has the best overall survival (~55-65% at 5 years) and is generally cured with intensive chemotherapy alone — high-dose cytarabine (HiDAC) consolidation — WITHOUT allogeneic transplant in first complete remission. The favorable group comprises core-binding-factor AML [t(8;21) and inv(16)/t(16;16)], NPM1-mutated without FLT3-ITD, and in-frame bZIP CEBPA mutation.

BOARD TRAP

Favorable risk means you AVOID allo-HCT in CR1 — transplant exposes a curable patient to needless transplant-related mortality. Risk is set by genetics, not blast count.

2. CBF forge

WHAT YOU SEE

The left station signed CBF FORGE with an anvil — the core-binding-factor smithy forges the t(8;21) and inv(16) fusions that disrupt the CBF complex.

WHAT IT ENCODES

Core-binding-factor (CBF) AML = t(8;21) RUNX1::RUNX1T1 and inv(16)/t(16;16) CBFB::MYH11. Both disrupt the CBF transcription complex (RUNX1 + CBFB) needed for normal hematopoiesis. These are AML-DEFINING genetic abnormalities — they are AML regardless of blast count (the ≥20% blast threshold does not apply).

BOARD TRAP

CBF AML (t8;21 and inv16) is diagnostic of AML even with <20% blasts; do not require 20% blasts when an AML-defining fusion is present.

3. Auer rod sword t(8;21)

WHAT YOU SEE

A sword-shaped Auer rod stamped 8;21 on the anvil (RUNX1-RUNX1T1) — the bladed crystal is the Auer-rod-rich, favorable t(8;21) fusion.

WHAT IT ENCODES

t(8;21)(q22;q22) creates the RUNX1::RUNX1T1 fusion, classically with abundant Auer rods and MPO positivity, and is favorable risk. CBF AML benefits from adding gemtuzumab ozogamicin (anti-CD33) to induction.

BOARD TRAP

The t(8;21) RUNX1::RUNX1T1 FUSION is favorable, whereas a RUNX1 point MUTATION is adverse — same gene name, opposite prognosis. Also, adverse KIT mutations can co-occur with CBF AML and worsen its outcome.

4. inv(16) eosinophils

WHAT YOU SEE

Pink abnormal eosinophils clustered by the anvil — the granular eosinophils are the M4Eo marrow clue to favorable inv(16)/CBFB-MYH11.

WHAT IT ENCODES

inv(16)(p13q22) or t(16;16) yields CBFB::MYH11 and is favorable; the marrow shows characteristic abnormal eosinophils with basophilic granules (AML M4Eo / acute myelomonocytic leukemia with eosinophilia).

BOARD TRAP

Abnormal marrow eosinophilia points to inv(16) CBF AML; like t(8;21), a co-occurring KIT mutation can downgrade the otherwise favorable prognosis.

5. NPM1 crown

WHAT YOU SEE

A crowned king at the center NPM1 station holding a crown (CROWN FORGE) — the regal crown is mutated nucleophosmin, favorable only without FLT3-ITD.

WHAT IT ENCODES

NPM1 (nucleophosmin) mutation is one of the most common AML lesions (~30%), typically with normal karyotype, and is FAVORABLE only in the absence of FLT3-ITD. It is a strong target for measurable residual disease monitoring by PCR.

BOARD TRAP

NPM1 is favorable ONLY when FLT3-ITD is absent; NPM1 + FLT3-ITD is intermediate risk, not favorable.

6. Cytoplasmic NPM1

WHAT YOU SEE

The king's crown displaced down out of place (cytoplasmic) — the misplaced crown is aberrant cytoplasmic NPM1 localization, the IHC clue (NPM1c+).

WHAT IT ENCODES

Mutated NPM1 carries a frameshift in exon 12 that creates a nuclear export signal, causing aberrant CYTOPLASMIC localization of nucleophosmin (normally nucleolar) — detectable by immunohistochemistry (NPM1c+).

BOARD TRAP

Cytoplasmic NPM1 staining is the morphologic clue to NPM1 mutation; it is AML-defining and persists at relapse, making it ideal for MRD tracking.

7. FLT3-ITD red chain

WHAT YOU SEE

A heavy red FLT3 chain padlocking the FAVORABLE sign shut — the chain that 'blocks' favorable status is FLT3-ITD downgrading NPM1 to intermediate.

WHAT IT ENCODES

A concurrent FLT3-ITD removes NPM1's favorable status, downgrading it to intermediate risk in ELN 2022 (regardless of allelic ratio). These patients add midostaurin to 7+3 and are considered for allo-HCT.

BOARD TRAP

NPM1-mut + FLT3-ITD is NOT favorable — it is intermediate; the FLT3-ITD overrides the favorable NPM1. (In ELN 2017 the allelic ratio mattered; in ELN 2022 any FLT3-ITD with NPM1 is intermediate.)

8. CEBPA bZIP forge

WHAT YOU SEE

The right station signed CEBPA ZIPPER FORGE with a zipper motif — the bZIP zipper is the in-frame CEBPA mutation that confers favorable risk.

WHAT IT ENCODES

An in-frame mutation in the basic leucine zipper (bZIP) region of CEBPA is favorable. ELN 2022 narrowed the definition: only bZIP in-frame CEBPA mutations qualify (whether mono- or biallelic), replacing the older biallelic-CEBPA requirement.

BOARD TRAP

Only bZIP in-frame CEBPA confers favorable risk (ELN 2022); non-bZIP CEBPA mutations do not. The older rule required BIALLELIC CEBPA — both are testable nuances.

9. ~55-60% survival

WHAT YOU SEE

Big percentage signs reading 55%, 60% (and 30%) hung over the forges — the high survival numbers are why favorable AML gets chemo, not transplant.

WHAT IT ENCODES

Favorable cytogenetics/genetics carry the highest long-term survival in AML (~55-65% at 5 years), which is precisely why chemotherapy consolidation alone — not transplant — is the standard in CR1.

BOARD TRAP

High survival is contingent on appropriate consolidation (HiDAC) AND on MRD clearance; persistent MRD reclassifies the patient toward transplant.

10. MRD ticker tape

WHAT YOU SEE

A PCR ticker tape labeled MRD spooling out '2 CYCLES' — the printout is molecular residual-disease monitoring after ~2 cycles that can override 'no transplant'.

WHAT IT ENCODES

NPM1 (and the CBF fusions) provide molecular targets for measurable residual disease (MRD) monitoring by quantitative PCR, typically after ~2 cycles and through follow-up. MRD positivity/rising transcripts predict relapse and can override the 'favorable, no transplant' default.

BOARD TRAP

A favorable-risk patient who is MRD-positive after consolidation is no longer treated as low-risk — persistent/rising MRD is an indication to consider allo-HCT.

11. No HCT sign

WHAT YOU SEE

A crossed-bone sign reading NO HCT at the base — the struck-out transplant icon means favorable-risk patients skip allo-HCT in CR1.

WHAT IT ENCODES

Favorable-risk patients in CR1 generally do NOT receive allogeneic transplant; they are consolidated with HiDAC, because transplant's treatment-related mortality outweighs benefit in a group cured by chemo.

BOARD TRAP

Don't transplant favorable-risk AML in first remission by default — reserve allo-HCT for relapse (CR2) or MRD failure.

12. Myeloid sarcoma

WHAT YOU SEE

A glowing green crystal 'spine/mass' (myeloid sarcoma) — the green chloroma is an extramedullary myeloblast tumor that is itself AML-defining.

WHAT IT ENCODES

Myeloid (granulocytic) sarcoma — an extramedullary tumor of myeloblasts (chloroma, green from MPO) — can occur with t(8;21) CBF AML. It is itself AML-defining: tissue diagnosis of myeloid sarcoma equals AML regardless of marrow blast percentage.

BOARD TRAP

A myeloid sarcoma (chloroma) IS acute leukemia even if the marrow shows <20% blasts; treat as systemic AML, not as an isolated tumor.

13. CD33 / gemtuzumab gemstone

WHAT YOU SEE

A blue gemstone marked 33 with an antibody beside the eosinophils — the targeted gem is CD33, bound by gemtuzumab (anti-CD33 ADC) in favorable/CBF AML.

WHAT IT ENCODES

CD33 is expressed on most AML myeloblasts. Gemtuzumab ozogamicin is an anti-CD33 antibody-drug conjugate (delivers calicheamicin) added to induction; it improves survival particularly in CD33+ favorable/CBF AML. Watch for veno-occlusive disease (sinusoidal obstruction syndrome).

BOARD TRAP

Gemtuzumab's benefit is greatest in favorable/CBF (CD33+) AML and least in adverse-risk; key toxicity is hepatic veno-occlusive disease/SOS.